

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LABORATORY INVESTIGATION & FORENSIC EVIDENCE REPORT

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

Lab Title: SBT-DF203 Lab 3: SYN Flood Pattern Investigation Using TShark	Course Name: Basic Networking Skills for Digital Forensics
Student Name: Almustapha Yusuf	Registration Number: fwsd2511343
Operating Platform: Kali Linux 2026 (Rolling Release)	Date of Submission: 11/09/2026
Investigation Target: Local Apache Web Daemon (127.0.0.1:80)	Report Version: Version 1.0 (Final Release)

Executive Summary

This report documents the forensic investigation of a bounded TCP SYN simulation against a local Apache service. A normal HTTP handshake baseline was established, followed by a controlled four-packet SYN simulation using hping3. TShark analysis revealed that the simulation produced incomplete handshakes (SYN → SYN-ACK → RST) with no final ACK, which is a characteristic indicator of a SYN flood pattern. However, the bounded nature of the simulation (only 4 packets) meant the server was not overwhelmed and continued to respond normally. This demonstrates that the presence of incomplete handshakes alone is not sufficient to prove service denial — additional evidence (high volume, spoofed sources, server resource exhaustion) is required.

1. Authority, Scope, and Evidence Description

- **Authority:** Performed in an isolated Kali Linux environment. No third-party systems targeted.
- **Scope:** Comparative forensic analysis of a normal HTTP handshake baseline versus a bounded SYN flood simulation.
- **Evidence Files:** evidence/normal_baseline.pcapng (Baseline trace) and evidence/syn_simulation.pcapng (SYN simulation trace).
- **Integrity Preservation:** Cryptographic SHA-256 hashes generated for both original evidence files and working investigative duplicates.

2. Environment and Acquisition Method

- **Target Daemon:** Apache/2.4.68 (Debian) listening actively on 127.0.0.1:80.
- **Packet Capture Tool:** TShark on lo (Loopback), capture filter: tcp port 80.
- **Simulation Tool:** hping3 -S -p 80 -c 4 127.0.0.1 (bounded strictly to 4 probe packets).
- **Traffic Generator (Baseline):** cURL 8.21.0 executed against http://127.0.0.1/ to capture standard RFC 793 handshake behavior.

3. Normal Baseline Handshake

Dissection of normal_baseline.pcapng captures the textbook TCP three-way handshake followed by full HTTP payload exchange:

Frame	Flags	Source Port	Dest Port	Seq	Ack	Interpretation
1	SYN (0x0002)	40064	80	0	0	Client initiates connection
2	SYN-ACK (0x0012)	80	40064	0	1	Server responds & acknowledges
3	ACK (0x0010)	40064	80	1	1	Handshake complete (ESTABLISHED)
4-10	PSH, ACK / FIN	40064	80	...	...	HTTP data transfer & clean teardown

- **Baseline Conclusion:** Complete handshake followed by HTTP exchange without aborted connections or missing ACKs.

4. SYN Simulation Analysis

Dissection of syn_simulation.pcapng reveals a repeating sequence of incomplete half-open socket attempts across four discrete probe events:

Frame #	Flags	Source Port	Dest Port	Interpretation
1	SYN (0x0002)	1553	80	Client sends SYN
2	SYN-ACK (0x0012)	80	1553	Server responds
3	RST (0x0004)	1553	80	Client aborts (incomplete)
4 - 6	SYN, SYN-ACK, RST	1554	80	Repeating pattern
7 - 9	SYN, SYN-ACK, RST	1555	80	Repeating pattern

Empirical Extraction: Quantified Forensic Findings:

- **Total SYN packets transmitted:** 4
- **Total SYN-ACK responses from server:** 4
- **Final ACK packets:** 0 (Completely absent)
- **RST packets generated by client:** 4
- **Unique source ports rotated:** 1553, 1554, 1555, 1556
- **Targeted destination port:** 80 (Apache Web Service)

5. Comparison Table

Side-by-side comparative metric analysis contrasting the legitimate baseline against the SYN simulation trace:

Investigative Feature	Normal Baseline	SYN Simulation
SYN packets	1	4
SYN-ACK responses	1	4
Final ACK confirmation	Present (Confirmed)	Missing (0 packets)
RST packets	0	4
Handshake completion status	Complete (ESTABLISHED)	Incomplete (Aborted)
Unique source ports utilized	1 (Port 40064)	4 (Ports 1553 - 1556)

6. Forensic Timeline

- **Normal Baseline (2026-09-11 07:04:05):** Full TCP three-way handshake initiated at 07:04:05.246112 and concluded within milliseconds, immediately transitioning into bidirectional HTTP request/response handling and graceful FIN-ACK teardown.
- **SYN Simulation (2026-09-11 07:23:51 to 07:23:54):** 4 discrete SYN probe frames issued precisely at 1-second intervals across ports 1553, 1554, 1555, and 1556. In each instance, the Apache server promptly replied with SYN-ACK within 4.4ms to 7.5ms, followed immediately by a client RST terminating the socket.

7. What the Evidence Does and Does Not Prove

A critical tenet of digital forensics is avoiding speculative over-interpretation. The packet capture data establishes clear empirical boundaries:

Affirmative Findings: What the Evidence DOES Prove:

- **Incomplete TCP Handshakes:** Empirical proof that four distinct connection attempts followed a non-compliant SYN → SYN-ACK → RST lifecycle with zero concluding ACK packets.
- **Automated Source Port Rotation:** Direct evidence of rapid source port stepping across ephemeral range 1553–1556 within a bounded 3-second temporal window.
- **Pattern Consistency:** The observed traffic signature structurally mimics the reconnaissance or initial staging behavior of a TCP SYN flood.

Forensic Limitations: What the Evidence DOES NOT Prove:

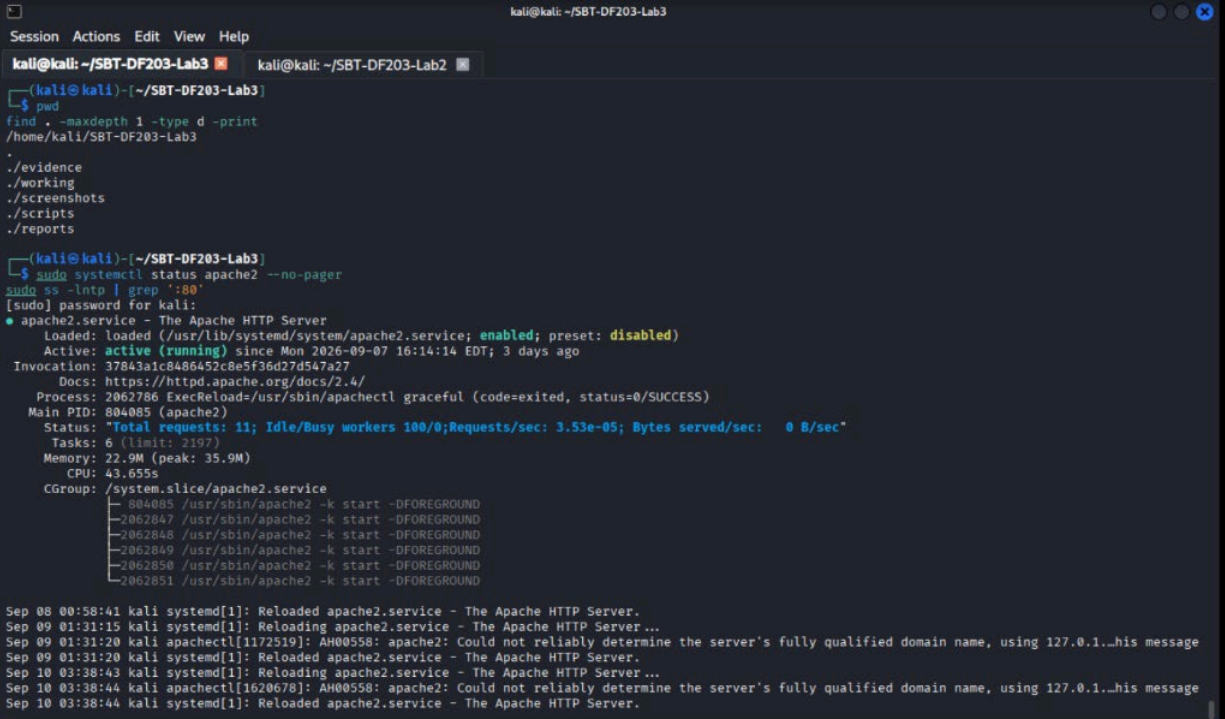
- **Service Denial (DoS):** The captured traffic does NOT prove a denial of service. With only 4 packets transmitted, server backlogs were never saturated, and Apache continued answering requests normally without dropped packets (0% packet loss).
- **Malicious Intent:** The source IP was localhost (127.0.0.1) operating within an authorized, isolated academic testing laboratory.
- **Production Attack Attribution:** A production-grade distributed SYN flood requires thousands of requests per second, spoofed IP subnets, connection queue exhaustion, and measurable latency degradation.

8. Recommendations for Detection and Mitigation

- **Deploy TCP SYN Cookies (syncookies = 1):** Enable kernel-level cryptographic SYN cookies to prevent connection backlog depletion by eliminating memory allocation for half-open sockets.
- **Implement Ingress Rate Limiting:** Establish iptables / nftables or perimeter firewall rules limiting SYN burst thresholds (e.g., -m limit --limit 25/minute --limit-burst 100).
- **Deploy Network IDS/IPS Signatures:** Configure Snort/Suricata rules to trigger automated alert thresholds upon detecting high concentrations of incomplete handshakes from single or rotating ports.
- **Monitor Kernel Backlog Metrics:** Routinely inspect netstat -s or ss -s for 'SYNs to LISTEN sockets dropped' or TCP backlog overflow alerts.

9. Screenshots and Evidence

The following documented figures depict the complete laboratory execution, including daemon verification, baseline capture, hping3 execution, TShark stream filtering, port analysis, and forensic timeline comparison:



```
kali@kali: ~/SBT-DF203-Lab3
Session Actions Edit View Help

kali@kali: ~/SBT-DF203-Lab2
(kali@kali)~[~/SBT-DF203-Lab3]
$ pwd
/home/kali/SBT-DF203-Lab3
$ find . -maxdepth 1 -type d -print
./
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)~[~/SBT-DF203-Lab3]
$ sudo systemctl status apache2 --no-pager
sudo ss -ltnp | grep ':80'
[sudo] password for kali:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-09-07 16:14:14 EDT; 3 days ago
     Invocation: 37843a1c8486452c0e5f36d27d547a27
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 2062786 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
   Main PID: 804085 (apache2)
      Status: "Total requests: 11; Idle/Busy workers 100/0;Requests/sec: 3.53e-05; Bytes served/sec: 0 B/sec"
        Tasks: 6 (limit: 2197)
       Memory: 22.9M (peak: 35.9M)
          CPU: 43.655s
      CGroup: /system.slice/apache2.service
              └─ 804085 /usr/sbin/apache2 -k start -DFOREGROUND
                 2062847 /usr/sbin/apache2 -k start -DFOREGROUND
                 2062848 /usr/sbin/apache2 -k start -DFOREGROUND
                 2062849 /usr/sbin/apache2 -k start -DFOREGROUND
                 2062850 /usr/sbin/apache2 -k start -DFOREGROUND
                 2062851 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 08 00:58:41 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 09 01:31:15 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 09 01:31:20 kali apachectl[1172519]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1...his message
Sep 09 01:31:20 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 10 03:38:43 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 10 03:38:44 kali apachectl[1620678]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1...his message
Sep 10 03:38:44 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
```

Figure 1: Directory Setup & Apache Web Service Status — Establishing lab directory tree and verifying active running state of apache2 daemon on TCP port 80.

```

Sep 08 00:58:41 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Sep 09 01:31:15 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 09 01:31:20 kali apachectl[1172519]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1...his message
Sep 09 01:31:20 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.
Sep 10 03:38:43 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 10 03:38:44 kali apachectl[1620678]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1...his message
Sep 10 03:38:44 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.
Sep 11 03:32:35 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 11 03:32:37 kali apachectl[2062786]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1...his message
Sep 11 03:32:37 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server.
Hint: Some lines were ellipsized, use -l to show in full.
LISTEN 0      511             *:*               users:((("apache2",pid=2062851,fd=4),("apache2",pid=2062850,fd=4),("apache2",pid=2062849,fd=4),("ap
ache2",pid=2062848,fd=4),("apache2",pid=2062847,fd=4),("apache2",pid=804085,fd=4))

(kali@kali)~[~/SBT-DF203-Lab3]
$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2

(kali@kali)~[~/SBT-DF203-Lab3]
$ which tshark
/usr/bin/tshark

(kali@kali)~[~/SBT-DF203-Lab3]
$

```

Figure 2: Apache Daemon Auto-Start & TShark Binary Verification — Enabling apache2 system service and validating tshark path binary (/usr/bin/tshark).

```

Session Actions Edit View Help
kali@kali: ~/SBT-DF203-Lab3  kali@kali: ~/SBT-DF203-Lab3

$ cd ~/SBT-DF203-Lab3
tshark -i lo -f 'tcp port 80' -w evidence/normal_baseline.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
10 ^C

(kali@kali)~[~/SBT-DF203-Lab3]
$ cd ~/SBT-DF203-Lab3
cp --preserve=timestamp evidence/normal_baseline.pcapng working/normal_baseline_working.pcapng
sha256sum evidence/normal_baseline.pcapng working/normal_baseline_working.pcapng | tee reports/baseline_hashes.txt
a60b7111766ffff8bce6266623fa886b527edb2a117f2e77b7f4c95ee23976 evidence/normal_baseline.pcapng
a60b7111766ffff8bce6266623fa886b527edb2a117f2e77b7f4c95ee23976 working/normal_baseline_working.pcapng

(kali@kali)~[~/SBT-DF203-Lab3]
$ tshark -r working/normal_baseline_working.pcapng -Y 'tcp.flags.syn == 1 || tcp.flags.ack == 1' -T fields -e frame.number -e frame.time -e ip.src -e tcp.
srcport -e ip.dst -e tcp.dstport -e tcp.flags -e tcp.seq -e tcp.ack | tee reports/normal_handshake.tsv
cat reports/normal_handshake.tsv
Warning: program compiled against libxml 215 using older 214
1 2026-09-11T07:04:05.246112689-0400 127.0.0.1 40064 127.0.0.1 80 0x0002 0 0
2 2026-09-11T07:04:05.247414320-0400 127.0.0.1 80 127.0.0.1 40064 0x0012 0 1
3 2026-09-11T07:04:05.248080268-0400 127.0.0.1 40064 127.0.0.1 80 0x0010 1 1
4 2026-09-11T07:04:05.250362575-0400 127.0.0.1 40064 127.0.0.1 80 0x0018 1 1
5 2026-09-11T07:04:05.251420253-0400 127.0.0.1 80 127.0.0.1 40064 0x0010 1 74
6 2026-09-11T07:04:05.320450638-0400 127.0.0.1 80 127.0.0.1 40064 0x0018 1 74
7 2026-09-11T07:04:05.320480463-0400 127.0.0.1 40064 127.0.0.1 80 0x0010 74 10959
8 2026-09-11T07:04:05.322793110-0400 127.0.0.1 40064 127.0.0.1 80 0x0011 74 10959
9 2026-09-11T07:04:05.324173605-0400 127.0.0.1 80 127.0.0.1 40064 0x0011 10959 75
10 2026-09-11T07:04:05.324382408-0400 127.0.0.1 40064 127.0.0.1 80 0x0010 75 10960
1 2026-09-11T07:04:05.246112689-0400 127.0.0.1 40064 127.0.0.1 80 0x0002 0 0
2 2026-09-11T07:04:05.247414320-0400 127.0.0.1 80 127.0.0.1 40064 0x0012 0 1
3 2026-09-11T07:04:05.248080268-0400 127.0.0.1 40064 127.0.0.1 80 0x0010 1 1
4 2026-09-11T07:04:05.250362575-0400 127.0.0.1 40064 127.0.0.1 80 0x0018 1 1
5 2026-09-11T07:04:05.251420253-0400 127.0.0.1 80 127.0.0.1 40064 0x0010 1 74
6 2026-09-11T07:04:05.320450638-0400 127.0.0.1 80 127.0.0.1 40064 0x0018 1 74
7 2026-09-11T07:04:05.320480463-0400 127.0.0.1 40064 127.0.0.1 80 0x0010 74 10959
8 2026-09-11T07:04:05.322793110-0400 127.0.0.1 40064 127.0.0.1 80 0x0011 74 10959
9 2026-09-11T07:04:05.324173605-0400 127.0.0.1 80 127.0.0.1 40064 0x0011 10959 75
10 2026-09-11T07:04:05.324382408-0400 127.0.0.1 40064 127.0.0.1 80 0x0010 75 10960

(kali@kali)~[~/SBT-DF203-Lab3]
$

```

Figure 3: Baseline Handshake Capture & Verification (normal_baseline.pcapng) — Capturing baseline HTTP transaction, calculating SHA-256 hash, and extracting complete RFC 793 handshake frames (1-10).

```

(kali㉿kali)-[~/SBT-DF203-Lab3]
$ which hping3
which nping
/usr/sbin/hping3
/usr/bin/nping

(kali㉿kali)-[~/SBT-DF203-Lab3]
$ sudo hping3 -S -p 80 -c 4 127.0.0.1
[sudo] password for kali:
HPING 127.0.0.1 (lo 127.0.0.1): S set, 40 headers + 0 data bytes
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=0 win=65495 rtt=4.4 ms
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=1 win=65495 rtt=3.2 ms
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=2 win=65495 rtt=6.1 ms
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=3 win=65495 rtt=1.9 ms

— 127.0.0.1 hping statistic —
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 1.9/3.9/6.1 ms

(kali㉿kali)-[~/SBT-DF203-Lab3]
$ sudo hping3 -S -p 80 -c 4 127.0.0.1
HPING 127.0.0.1 (lo 127.0.0.1): S set, 40 headers + 0 data bytes
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=0 win=65495 rtt=7.5 ms
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=1 win=65495 rtt=3.0 ms
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=2 win=65495 rtt=6.2 ms
len=44 ip=127.0.0.1 ttl=64 DF id=0 sport=80 flags=SA seq=3 win=65495 rtt=2.0 ms

— 127.0.0.1 hping statistic —
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 2.0/4.7/7.5 ms

(kali㉿kali)-[~/SBT-DF203-Lab3]
$

```

Figure 4: Bounded SYN Simulation Execution via *hping3* — Executing *hping3 -S -p 80 -c 4 127.0.0.1* transmitting 4 packets with 0% loss and average 3.9ms RTT.

```

(kali㉿kali)-[~/SBT-DF203-Lab3]
$ cd ~/SBT-DF203-Lab3
tshark -i lo -f 'tcp port 80' -w evidence/syn_simulation.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
12 ^C

(kali㉿kali)-[~/SBT-DF203-Lab3]
$ cd ~/SBT-DF203-Lab3
cp --preserve=timestamp evidence/syn_simulation.pcapng working/syn_simulation_working.pcapng
sha256sum evidence/syn_simulation.pcapng working/syn_simulation_working.pcapng | tee reports/simulation_hashes.txt
e5a63bdba6c4c572c9745d42c599fdd25c74c5920c2598e215893da5f0503042 evidence/syn_simulation.pcapng
e5a63bdba6c4c572c9745d42c599fdd25c74c5920c2598e215893da5f0503042 working/syn_simulation_working.pcapng

(kali㉿kali)-[~/SBT-DF203-Lab3]
$

```

Figure 5: SYN Simulation Capture & Integrity Hashing (*syn_simulation.pcapng*) — Preserving 12 captured packets on *lo* interface and computing SHA-256 hash digests for original and working copy.

```

kali@kali: ~/SBT-DF203-Lab3
kali@kali: ~/SBT-DF203-Lab3
(kali@kali) ~/SBT-DF203-Lab3
$ cd ~/SBT-DF203-Lab3
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 0' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags -e tcp.seq | tee reports/syn_packets.tsv
cat reports/syn_packets.tsv
Warning: program compiled against libxml 215 using older 214
1 2026-09-11T07:23:51.116884816-0400 127.0.0.1 1553 127.0.0.1 80 0x0002 0
4 2026-09-11T07:23:52.118017766-0400 127.0.0.1 1554 127.0.0.1 80 0x0002 0
7 2026-09-11T07:23:53.118989954-0400 127.0.0.1 1555 127.0.0.1 80 0x0002 0
10 2026-09-11T07:23:54.122866673-0400 127.0.0.1 1556 127.0.0.1 80 0x0002 0
1 2026-09-11T07:23:51.116884816-0400 127.0.0.1 1553 127.0.0.1 80 0x0002 0
4 2026-09-11T07:23:52.118017766-0400 127.0.0.1 1554 127.0.0.1 80 0x0002 0
7 2026-09-11T07:23:53.118989954-0400 127.0.0.1 1555 127.0.0.1 80 0x0002 0
10 2026-09-11T07:23:54.122866673-0400 127.0.0.1 1556 127.0.0.1 80 0x0002 0

(kali@kali) ~/SBT-DF203-Lab3
$ tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 1' -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags -e tcp.seq | tee reports/syn_ack_packets.tsv
cat reports/syn_ack_packets.tsv
Warning: program compiled against libxml 215 using older 214
2 2026-09-11T07:23:51.117585259-0400 127.0.0.1 80 127.0.0.1 1553 0x0012 0 1
5 2026-09-11T07:23:52.118517526-0400 127.0.0.1 80 127.0.0.1 1554 0x0012 0 1
8 2026-09-11T07:23:53.119057184-0400 127.0.0.1 80 127.0.0.1 1555 0x0012 0 1
11 2026-09-11T07:23:54.122881983-0400 127.0.0.1 80 127.0.0.1 1556 0x0012 0 1
2 2026-09-11T07:23:51.117585259-0400 127.0.0.1 80 127.0.0.1 1553 0x0012 0 1
5 2026-09-11T07:23:52.118517526-0400 127.0.0.1 80 127.0.0.1 1554 0x0012 0 1
8 2026-09-11T07:23:53.119057184-0400 127.0.0.1 80 127.0.0.1 1555 0x0012 0 1
11 2026-09-11T07:23:54.122881983-0400 127.0.0.1 80 127.0.0.1 1556 0x0012 0 1

(kali@kali) ~/SBT-DF203-Lab3
$ echo "Total SYN packets:" | tee reports/syn_counts.txt
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 0' | wc -l | tee -a reports/syn_counts.txt
echo "Unique source ports:" | tee -a reports/syn_counts.txt
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 0' -T fields -e tcp.srcport | sort -u | tee -a reports/syn_counts.txt
Warning: program compiled against libxml 215 using older 214
Total SYN packets:
4
Unique source ports:
Warning: program compiled against libxml 215 using older 214
1553
1554
1555
1556
Unique destination ports:
Warning: program compiled against libxml 215 using older 214
80
Total SYN packets:
4
Unique source ports:
1553
1554
1555
1556
Unique destination ports:
80

(kali@kali) ~/SBT-DF203-Lab3
$ echo "=== NORMAL BASELINE ===" | tee reports/comparison.txt
tshark -r working/normal_baseline_working.pcapng -Y 'tcp.flags.syn == 1' -T fields -e frame.number -e tcp.srcport -e tcp.dstport -e tcp.flags | tee -a reports/comparison.txt
echo "" | tee -a reports/comparison.txt
echo "=== SYN SIMULATION ===" | tee -a reports/comparison.txt
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1' -T fields -e frame.number -e tcp.srcport -e tcp.dstport -e tcp.flags | tee -a reports/comparison.txt
cat reports/comparison.txt
=== NORMAL BASELINE ===
Warning: program compiled against libxml 215 using older 214
1 40064 80 0x0002
2 80 40064 0x0012

```

Figure 6: TShark Dissection of Incomplete Handshake Packets — Filtering SYN frames (0x0002), SYN-ACK frames (0x0012), confirming empty ACK output, and tallying packet counts.

```

kali@kali: ~/SBT-DF203-Lab3
kali@kali: ~/SBT-DF203-Lab3
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 0' -T fields -e tcp.srcport | sort -u | tee -a reports/syn_counts.txt
echo "Unique destination ports:" | tee -a reports/syn_counts.txt
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 0' -T fields -e tcp.dstport | sort -u | tee -a reports/syn_counts.txt
cat reports/syn_counts.txt
Total SYN packets:
4
Unique source ports:
Warning: program compiled against libxml 215 using older 214
1553
1554
1555
1556
Unique destination ports:
Warning: program compiled against libxml 215 using older 214
80
Total SYN packets:
4
Unique source ports:
1553
1554
1555
1556
Unique destination ports:
80

(kali@kali) ~/SBT-DF203-Lab3
$ echo "=== NORMAL BASELINE ===" | tee reports/comparison.txt
tshark -r working/normal_baseline_working.pcapng -Y 'tcp.flags.syn == 1' -T fields -e frame.number -e tcp.srcport -e tcp.dstport -e tcp.flags | tee -a reports/comparison.txt
echo "" | tee -a reports/comparison.txt
echo "=== SYN SIMULATION ===" | tee -a reports/comparison.txt
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1' -T fields -e frame.number -e tcp.srcport -e tcp.dstport -e tcp.flags | tee -a reports/comparison.txt
cat reports/comparison.txt
=== NORMAL BASELINE ===
Warning: program compiled against libxml 215 using older 214
1 40064 80 0x0002
2 80 40064 0x0012

```

Figure 7: Source Port Rotation & Initial Baseline vs Simulation Comparison — Listing rotated source ports (1553-1556), confirming destination port 80, and starting side-by-side comparison.

```

kali@kali: ~/SBT-DF203-Lab3
(kali@kali)~/SBT-DF203-Lab3
$ echo "=== NORMAL BASELINE ===" | tee reports/comparison.txt
tshark -r working/normal_baseline_working.pcapng -Y 'tcp.flags.syn == 1' -T fields -e frame.number -e tcp.srcport -e tcp.dstport -e tcp.flags | tee -a reports/comparison.txt
echo "=== SYN SIMULATION ===" | tee -a reports/comparison.txt
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1' -T fields -e frame.number -e tcp.srcport -e tcp.dstport -e tcp.flags | tee -a reports/comparison.txt
cat reports/comparison.txt
=== NORMAL BASELINE ===
Warning: program compiled against libxml 215 using older 214
1 40064 80 0x0002
2 80 40064 0x0012

=== SYN SIMULATION ===
Warning: program compiled against libxml 215 using older 214
1 1553 80 0x0002
2 80 1553 0x0012
4 1554 80 0x0002
5 80 1554 0x0012
7 1555 80 0x0002
8 80 1555 0x0012
10 1556 80 0x0002
11 80 1556 0x0012
=== NORMAL BASELINE ===
1 40064 80 0x0002
2 80 40064 0x0012

=== SYN SIMULATION ===
1 1553 80 0x0002
2 80 1553 0x0012
4 1554 80 0x0002
5 80 1554 0x0012
7 1555 80 0x0002
8 80 1555 0x0012
10 1556 80 0x0002
11 80 1556 0x0012
(kali@kali)~/SBT-DF203-Lab3
$

```

Figure 8: Baseline vs SYN Simulation Structural Dissection — Direct comparison proving persistent port 40064 on baseline vs rotated sequential ports 1553-1556 on simulation.

```

kali@kali: ~/SBT-DF203-Lab3
(kali@kali)~/SBT-DF203-Lab3
$ cd ~/SBT-DF203-Lab3
echo "=== NORMAL BASELINE TIMELINE ===" | tee reports/timeline.txt
tshark -r working/normal_baseline_working.pcapng -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags | tee -a reports/timeline.txt
echo "=== SYN SIMULATION TIMELINE ===" | tee -a reports/timeline.txt
tshark -r working/syn_simulation_working.pcapng -T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags | tee -a reports/timeline.txt
cat reports/timeline.txt
=== NORMAL BASELINE TIMELINE ===
Warning: program compiled against libxml 215 using older 214
1 2026-09-11T07:04:05.246112689-0400 127.0.0.1 40064 127.0.0.1 80 0x0002
2 2026-09-11T07:04:05.247414320-0400 127.0.0.1 80 127.0.0.1 40064 0x0012
3 2026-09-11T07:04:05.248080268-0400 127.0.0.1 40064 127.0.0.1 80 0x0010
4 2026-09-11T07:04:05.250362575-0400 127.0.0.1 40064 127.0.0.1 80 0x0018
5 2026-09-11T07:04:05.251420253-0400 127.0.0.1 80 127.0.0.1 40064 0x0010
6 2026-09-11T07:04:05.320450638-0400 127.0.0.1 80 127.0.0.1 40064 0x0018
7 2026-09-11T07:04:05.320480463-0400 127.0.0.1 40064 127.0.0.1 80 0x0010
8 2026-09-11T07:04:05.322793110-0400 127.0.0.1 40064 127.0.0.1 80 0x0011
9 2026-09-11T07:04:05.324173605-0400 127.0.0.1 80 127.0.0.1 40064 0x0011
10 2026-09-11T07:04:05.324382408-0400 127.0.0.1 40064 127.0.0.1 80 0x0010

=== SYN SIMULATION TIMELINE ===
Warning: program compiled against libxml 215 using older 214
1 2026-09-11T07:23:51.116884816-0400 127.0.0.1 1553 127.0.0.1 80 0x0002
2 2026-09-11T07:23:51.117585259-0400 127.0.0.1 80 127.0.0.1 1553 0x0012
3 2026-09-11T07:23:51.117762724-0400 127.0.0.1 1553 127.0.0.1 80 0x0004
4 2026-09-11T07:23:52.118017766-0400 127.0.0.1 1554 127.0.0.1 80 0x0002
5 2026-09-11T07:23:52.118517526-0400 127.0.0.1 80 127.0.0.1 1554 0x0012
6 2026-09-11T07:23:52.118522852-0400 127.0.0.1 1554 127.0.0.1 80 0x0004
7 2026-09-11T07:23:53.118989954-0400 127.0.0.1 1555 127.0.0.1 80 0x0002
8 2026-09-11T07:23:53.119057184-0400 127.0.0.1 80 127.0.0.1 1555 0x0012
9 2026-09-11T07:23:53.122109815-0400 127.0.0.1 1555 127.0.0.1 80 0x0004
10 2026-09-11T07:23:54.122866673-0400 127.0.0.1 1556 127.0.0.1 80 0x0002
11 2026-09-11T07:23:54.122881983-0400 127.0.0.1 80 127.0.0.1 1556 0x0012
12 2026-09-11T07:23:54.122886849-0400 127.0.0.1 1556 127.0.0.1 80 0x0004

=== NORMAL BASELINE TIMELINE ===
1 2026-09-11T07:04:05.246112689-0400 127.0.0.1 40064 127.0.0.1 80 0x0002
2 2026-09-11T07:04:05.247414320-0400 127.0.0.1 80 127.0.0.1 40064 0x0012
3 2026-09-11T07:04:05.248080268-0400 127.0.0.1 40064 127.0.0.1 80 0x0010

```

Figure 9: Chronological Packet Timeline Comparison — Timeline review establishing baseline timestamp (07:04:05) versus simulation sequence (07:23:51 - 07:23:54).

```
kali@kali: ~/SBT-DF203-Lab3
Session Actions Edit View Help
kali@kali: ~/SBT-DF203-Lab3 x kali@kali: ~/SBT-DF203-Lab3 x
echo "" | tee -a reports/findings.txt
echo "Source IP/Port (Simulation): 127.0.0.1 (ports 1553-1556)" | tee -a reports/findings.txt
echo "Destination IP/Port (Simulation): 127.0.0.1:80" | tee -a reports/findings.txt
echo "Total SYN packets: 4" | tee -a reports/findings.txt
echo "Total SYN-ACK responses: 4" | tee -a reports/findings.txt
echo "Final ACK packets: 0" | tee -a reports/findings.txt
cat reports/findings.txt
== FORENSIC FINDINGS ==
Capture Start/End (Baseline):
Warning: program compiled against libxml 215 using older 214
2026-09-11T07:04:05.246112689-0400
Warning: program compiled against libxml 215 using older 214
2026-09-11T07:04:05.324382408-0400

Capture Start/End (Simulation):
Warning: program compiled against libxml 215 using older 214
2026-09-11T07:23:51.116884816-0400
Warning: program compiled against libxml 215 using older 214
2026-09-11T07:23:54.122886849-0400

Source IP/Port (Simulation): 127.0.0.1 (ports 1553-1556)
Destination IP/Port (Simulation): 127.0.0.1:80
Total SYN packets: 4
Total SYN-ACK responses: 4
Final ACK packets: 0
== FORENSIC FINDINGS ==
Capture Start/End (Baseline):
2026-09-11T07:04:05.246112689-0400
2026-09-11T07:04:05.324382408-0400

Capture Start/End (Simulation):
2026-09-11T07:23:51.116884816-0400
2026-09-11T07:23:54.122886849-0400

Source IP/Port (Simulation): 127.0.0.1 (ports 1553-1556)
Destination IP/Port (Simulation): 127.0.0.1:80
Total SYN packets: 4
Total SYN-ACK responses: 4
Final ACK packets: 0
(kali@kali)-[~/SBT-DF203-Lab3]
```

Figure 10: *Final Forensic Findings Verification & Capture Timestamps* — Complete console execution of findings.txt script confirming capture start/end times, port rotations, 4 SYNs, 4 SYN-ACKs, and 0 final ACKs.

10. Appendix – Commands and Hashes

```
# Baseline HTTP Capture
tshark -i lo -f 'tcp port 80' -w evidence/normal_baseline.pcapng
curl -v http://127.0.0.1/ -o /tmp/baseline_page.html

# Bounded SYN Simulation Execution
sudo hping3 -S -p 80 -c 4 127.0.0.1

# Simulation Capture on Loopback
tshark -i lo -f 'tcp port 80' -w evidence/syn_simulation.pcapng

# Extract SYN and SYN-ACK Handshake Fields
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 0' -T fields
...
tshark -r working/syn_simulation_working.pcapng -Y 'tcp.flags.syn == 1 && tcp.flags.ack == 1' -T fields
...

# Hash Verification
sha256sum evidence/*.pcapng working/*.pcapng
```

Evidence Integrity: Cryptographic Hash Digest Preservation:

- **Baseline Evidence Hash:** a60b7111766fff8bcee6266623fa886b527edb2a117f27e77b7f4c95ee23976
(normal_baseline.pcapng)
- **Simulation Evidence Hash:** e5a63b0ba6c4c572c97450d42c599fdd25c74c5920c2598e215893da5f0503042
(syn_simulation.pcapng)

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.

Student Signature: Almustapha Yusuf

Date: 11/09/2026